

MAT1002: Calculus II L02

Quiz 2 — 2025-2026, Term 2

Wednesday, 4 Mar 2026

1. (a) (5 points) Express the series

$$\sum_{n=0}^{\infty} \frac{(-1)^n x^{n+1}}{n+1},$$

in the form of an elementary function. Identify its domain.

- (b) (10 points) Show that the Maclaurin series of $f(x) := \sin x$ converges to $f(x)$ for all $x \in \mathbb{R}$.

Solution:

- (a) By Ratio Test,

$$\left| \frac{a_{n+1}}{a_n} \right| = \left(\frac{n+1}{n+2} \right) |x| \rightarrow |x|,$$

as $n \rightarrow \infty$. The series converges when $|x| < 1$ and diverges for $|x| > 1$. Let the function f be defined as $f(x) := \sum (-1)^n x^{n+1} / (n+1)$ for $-1 < x < 1$. Consider its derivative

$$f'(x) = \sum_{n=0}^{\infty} (-1)^n x^n = \sum_{n=0}^{\infty} (-x)^n = \frac{1}{1+x},$$

for $|x| < 1$. Then

$$f(x) = \int f'(x) dx = \int \frac{1}{1+x} dx = \ln |1+x| + C,$$

where C is a constant. Using the fact that $0 = f(0) = \ln 1 + C = C$, we have $f(x) = \ln(1+x)$, $x \in (-1, 1)$. Note that it is okay to include $x = 1$ (No penalty if students do not include endpoints).

- (b) The function $f(x) = \sin x$ has derivatives of all orders for $x \in \mathbb{R}$. Note that

$$f'(x) = \cos(x), f''(x) = -\sin x, f'''(x) = -\cos x \implies f^{(2k)}(0) = 0, f^{(2k+1)}(0) = (-1)^k, k \in \mathbb{N}.$$

By Taylor's theorem, we write

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots + \frac{(-1)^k x^{2k+1}}{(2k+1)!} + R_{2k+1}(x),$$

where $R_{2k+1}(x)$ is a remainder. Since $|\cos(x)| \leq 1$, we have

$$|R_{2k+1}(x)| \leq 1 \cdot \frac{|x|^{2k+2}}{(2k+2)!} \rightarrow 0$$

as $n \rightarrow \infty$ and the series converges to $\sin x$ for every x . Thus,

$$\sin x = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k+1}}{(2k+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$$

2. (5 points) Find the limit

$$\lim_{x \rightarrow 0} \frac{\cos x - 1 + \frac{1}{2}x \sin x}{[\ln(1+x)]^4}.$$

Solution: We have

$$\begin{aligned} & \lim_{x \rightarrow 0} \frac{\cos x - 1 + \frac{1}{2}x \sin x}{[\ln(1+x)]^4} \\ &= \lim_{x \rightarrow 0} \frac{1 - \frac{x^2}{2!} + \frac{x^4}{4!} + \mathcal{O}(x^6) - 1 + \frac{x}{2} \left(x - \frac{x^3}{3!} + \mathcal{O}(x^5) \right)}{[x + \mathcal{O}(x^2)]^4} \\ &= \lim_{x \rightarrow 0} \frac{\frac{x^4}{4!} - \frac{x^4}{3! \cdot 2} + \frac{x}{2} \mathcal{O}(x^5) + \mathcal{O}(x^6)}{[x + \mathcal{O}(x^2)]^4} \\ &= \lim_{x \rightarrow 0} \frac{\left(\frac{1}{4!} - \frac{1}{3! \cdot 2} \right) + \mathcal{O}(x^2)}{[1 + \mathcal{O}(x)]^4} \\ &= \frac{1}{4!} - \frac{1}{3! \cdot 2} \\ &= \frac{1}{24} - \frac{1}{12} \\ &= -\frac{1}{24}. \end{aligned}$$

3. (a) (5 points) Consider the plane curve given by $x = t^2, y = t^3 - 3t, t \in \mathbb{R}$. Express the 2nd order derivative d^2y/dx^2 in terms of t .
- (b) (5 points) Find the length of the parametric curve given by $x = \sin(2t), y = \cos(2t), t \in [0, 2\pi]$.

Solution:

- (a) We have

$$\frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{d(dy/dx)/dt}{dx/dt},$$

and

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{3(t-1)}{2t}.$$

Substituting dy/dx back into the 1st equation, we have

$$\frac{d^2y}{dx^2} = \frac{\frac{d}{dt} \left(\frac{3t-3}{2t} \right)}{2t} = \frac{3(t^2+1)}{4t^3}.$$

- (b) The curve is the unit circle, but traced twice. It is traced exactly once when t goes from 0 to π . So the length is

$$L = \int_0^\pi \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt = \int_0^\pi \sqrt{4\cos^2(2t) + 4\sin^2(2t)} dt = 2 \int_0^\pi dt = 2\pi.$$